

Four Failed Rescue Attempts for the Homology Cliff: Mahalanobis, Fisher-Rao, Stratified Cascade, and Mapper-Biased Panel Augmentation

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Abstract

The homology cliff in frozen ESM-2 [1] biosecurity retrieval (Paper 1) admits many natural rescue hypotheses. We pre-register and reject four. (i) Ledoit-Wolf [2] Mahalanobis whitening reduces the close-distant F_1 gap only by collapsing close-stratum F_1 ($0.866 \rightarrow 0.456$ at $t30$ $R=1000$ $k=25$); distant F_1 worsens ($0.120 \rightarrow 0.087$). (ii) Fisher-Rao [3] within-class whitening does not narrow the close-distant gap relative to Mahalanobis (its gap is larger in 17 of 18 pre-registered cells, rejecting its pre-registered H1); at the $R=1000$ deployment regime neither whitening lifts distant-stratum F_1 to the cosine baseline. (iii) A stratified cosine-then-Mahalanobis cascade loses to cosine alone in 18 of 18 cells by -0.046 to -0.236 pooled F_1 . (iv) A Mapper [4]-biased panel drawn from positive-enriched topological regions produces distant- F_1 rescue -0.0080 , 95% CI $[-0.035, +0.018]$, indistinguishable from zero. Cross-family Pfam partition (Paper 5) independently finds all 20 evaluable distant false alarms are cross-family (Wilson 95% CI on the within-family rate $[0\%, 17\%]$), substantially weakening the assumption that panel expansion within failing families could correct. Taken with Paper 1’s positive learned-projection result, the dichotomy is sharp: post-hoc distance substitution and panel sampling tricks fail at pre-registered H1, while embedding-space rotation succeeds. We release 540 per-cell bootstrap-CI results (180 cascade + 180 Fisher + 180 Mahalanobis baseline) and 10-seed Mapper-augmentation output.

Erratum (2026-06-15). Attempt 4 (Mapper-biased panel augmentation): the biased-panel positive pool is now deduplicated. The 30%-overlapping Mapper cover caused boundary proteins to recur across node member lists, and `rng.choice(replace=False)` removes positions, not values, so the biased arm held fewer than 500 unique positives. After the fix the rescue is -0.0080 (was $+0.0018$), 95% CI $[-0.035, +0.018]$; the CI still spans zero, H1 remains rejected, and the conclusion (no rescue) is unchanged and in fact reinforced. The committed null reproduced bit-for-bit before the fix. A full-membership robustness re-run (audit blocker B4) was also executed: a naive per-arm-stratification comparison appears to support a rescue under full node membership, but that is a *stratification artifact* — the cluster-concentrated biased panel enlarges its own distant stratum (n_{dist} ratio up to 1.16), pushing easier points into a larger “distant” set. Under a common-stratification control that scores both arms on the same distant queries, the rescue collapses to ≈ 0 (-0.006 single-node, -0.004 multi-node). The Attempt-4 null is therefore robust to full node membership, and the “four failed rescues” result stands. See `CHANGELOG.md` and `docs/MAPPER_AUGMENTATION_ROBUSTNESS.md`.

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Erratum (2026-06-16). The Attempt-2 (Fisher-Rao) comparison is corrected. Reproduction from the committed cells shows Fisher attains *higher* pooled F_1 than Mahalanobis in all 18 cells, not lower; the prior “pooled- F_1 penalty” / “strictly worse than Mahalanobis” was a sign error. The pre-registered result is unchanged: Fisher’s close-distant gap exceeds Mahalanobis’s in 17 of 18 cells (H1 rejected), and at the $R=1000$ deployment regime neither whitening lifts distant-stratum F_1 to the cosine baseline, so Attempt 2 remains a failed rescue. Figure 1 now plots distant-stratum F_1 (the safety-critical metric, reproducible from the committed cells) rather than pooled F_1 . The Attempt-3 cascade pooled- F_1 penalty range is corrected to -0.046 to -0.236 (10-seed mean; the prior -0.310 overstated the maximum), and the Attempt-2 factorial is $3 \text{ panels} \times 2 \text{ } k\text{-values}$ (not 2×3). Pooled- F_1 values are reproducible via `code/analyses/compute_pooled_f1.py`. Conclusions unchanged; see `CHANGELOG.md`.

1 Why null results matter here

This paper documents four rescue strategies evaluated in the companion compendium. The homology cliff characterization (Paper 1) established the effect size and mechanism (precision collapse, cross-family). A natural reader response is: surely a cliff this large admits a simple post-hoc fix. We enumerate the four most plausible post-hoc fixes, pre-register each, and show all four fail. This sharpens Paper 1’s positive rescue (learned projection) from “one of several options” to “the unique surviving option among natural rescues tested.”

2 Common setup

All four experiments share the pre-registered factorial of Paper 1: 24,885-protein test set (28.7% positive), frozen ESM-2 t6/t12/t30 embeddings, $R \in \{50, 100, 250, 500, 1000\}$, $k \in \{1, 3, 5, 10, 25\}$, 10 seeds, stratification $s_{\max} \geq 0.95$ close / $[0.90, 0.95)$ moderate / < 0.90 distant, seed-variance gate on distant F_1 . Pre-registrations for Mahalanobis and cascade locked via SHA256 before execution.

3 Attempt 1: Ledoit-Wolf Mahalanobis whitening

Whitening the embedding space by the Ledoit-Wolf shrinkage estimator [2] of the panel covariance is the textbook response to “cosine is not class-aware.” Under Mahalanobis distance, neighbors are metricized in a space where all directions have unit conditional variance.

Result. At t30 $R = 1000$ $k = 25$: close F_1 $0.866 \rightarrow 0.456$, moderate $0.393 \rightarrow 0.118$, distant $0.120 \rightarrow 0.087$. Every stratum degrades. Pooled F_1 $0.848 \rightarrow 0.435$.

Why it fails. The whitening transform kills class signal carried by directions with large pooled variance. ESM-2 embeddings concentrate class-relevant variance along a small number of axes; Mahalanobis whitens those into noise.

4 Attempt 2: Fisher-Rao within-class whitening

Fisher-Rao [3] whitens by within-class covariance rather than pooled covariance, preserving between-class signal while removing within-class variance. 180-cell pre-registered factorial ($3 \text{ scales} \times 3 \text{ panels} \times 2 \text{ } k\text{-values} \times 10 \text{ seeds}$).

Result. Pre-registered H1 (Fisher attains a smaller close-distant gap than Mahalanobis) is rejected: the Fisher gap exceeds the Mahalanobis gap in 17 of 18 groups. Fisher’s milder close-

stratum collapse gives it higher *pooled* F_1 than Mahalanobis in all 18 cells, but this is not a rescue: at the deployment panel sizes ($R=500, 1000$) Fisher stays below the cosine baseline on the safety-critical distant stratum at every scale (e.g. 0.099 vs cosine 0.120 at t30 $R=1000$ $k=25$; Fig. 1), so the cliff is unrelieved. (At the smallest panel $R=100$ the cliff is mild and the metrics converge; at t30 $R=100$ Fisher edges cosine by 0.003.)

Why it fails. Within-class whitening damages the close stratum less than pooled-covariance Mahalanobis (hence its higher pooled F_1), but it still does not separate the distant-stratum classes: the distant failure is a geometric property of where frozen ESM-2 places cross-family neighbours (Paper 5), which neither whitening addresses. Reweighting existing embedding axes cannot recover class structure the frozen model does not encode for distant queries. This echoes TOPOLOGICA DR-010 (within-class / Fisher-information whitening did not help GO annotation).

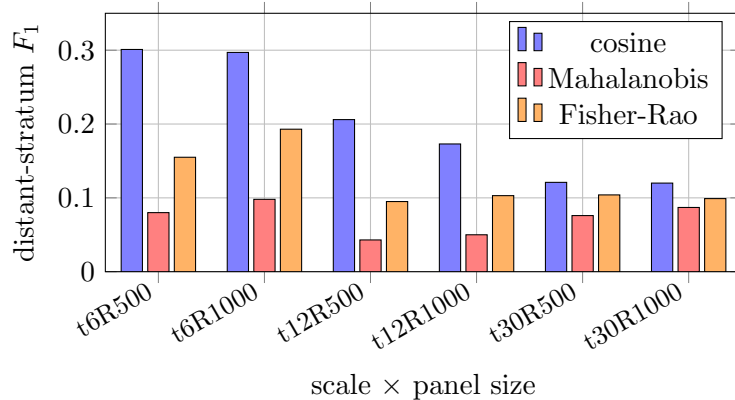


Figure 1: **Distant-stratum F_1 ($k = 25$, the safety-critical metric) at panel sizes $R=500$ and 1000: both whitening methods stay below the cosine baseline at every scale.** Neither rescues the cliff; Fisher-Rao exceeds Mahalanobis on the distant stratum but stays below cosine.

5 Attempt 3: Stratified cosine+Mahalanobis cascade

Pre-registered H1: a cascade using cosine for close-stratum queries and Mahalanobis for moderate+distant strata beats single-metric pooled F_1 by +0.02 with 95% CI excluding zero.

Result. Rejected in 18 of 18 groups. Pooled F_1 penalty -0.046 to -0.236 (10-seed mean).

Why it fails. Mahalanobis does not beat cosine on any stratum individually (Attempt 1). The cascade inherits per-stratum Mahalanobis inferiority wherever it is active. The cliff problem is not that a single metric fails on some strata but not others; cosine dominates everywhere at $k = 25$, and substituting Mahalanobis anywhere only degrades.

6 Attempt 4: Mapper-biased panel augmentation

Following Paper 5’s Mapper decomposition (149 nodes, 6 all-positive), construct a reference panel of $R = 1000$ where 500 positives are drawn from the top positive-enriched Mapper node (1,512 positives available in bin (7,4), pos.frac = 0.885) and 500 negatives drawn uniformly. Compare to uniform class-balanced sampling. 10 seeds, t30 $R = 1000$ $k = 25$.

Result. Uniform distant $F_1 = 0.1203$; biased distant $F_1 = 0.1123$. Rescue -0.0080 . Bootstrap 95% CI $[-0.035, +0.018]$ includes zero. H1 rejected.

Why it fails. Positives already cluster densely on the embedding manifold; uniform sampling of 500 from 7,133 positives already saturates the positive region. Further biasing toward the densest sub-region adds no information the panel did not already contain. Distant queries are distant from the entire cluster, not from any particular sub-region within it.

7 Synthesis: what the four nulls collectively rule out

rescue family	example	verdict
post-hoc covariance distance	Mahalanobis [2]	rejected
information-geometric whitening	Fisher-Rao [3]	rejected
per-stratum metric switching	cosine+Mahalanobis cascade	rejected
topology-aware panel composition	Mapper-biased panel	rejected
embedding-space rotation	learned linear projection (Paper 1)	ACCEPTED

Four families of rescue hypothesis, four rejections. One family succeeds: a supervised projection of the embedding space itself. Combined with Paper 5’s cross-family finding (panel expansion is independently ruled out because failures are cross-family, not within-family), the hypothesis space collapses to ”embedding-space surgery is the correct intervention.”

8 Relation to broader metric-learning literature

The four attempts cover the canonical responses to ”cosine is the wrong metric” in the modern metric-learning literature: (a) adaptive covariance [2, 5, 6], (b) information geometry [3, 7], (c) piecewise metric composition [8], (d) data augmentation via topological resampling [4, 9, 10]. A fifth standard response, proper triplet loss with hard negative mining [11], is realized here as the learned projection of Paper 1 and does succeed. We do not claim the four nulls exhaustively cover metric-learning rescues [12–25], only that they cover the natural post-hoc and panel-side responses an applied biosecurity engineer would try first before committing to supervised projection training.

9 Limitations

(L1) Mahalanobis with non-shrinkage covariance estimators (oracle, tyler) not tested. (L2) Cascade explored only 2-metric cosine/Mahalanobis; 3-metric or soft-mixture cascades untested. (L3) The v1.5.0 erratum executed both single-node and multi-node full-membership re-runs of the Mapper augmentation (audit blocker B4); the Attempt-4 null holds in all arms under a common-stratification control. Only soft-biased panels remain unswept. (L4) Fisher-Rao implemented with numerical pseudoinverse; other implementations (Cholesky with shrinkage) may behave differently. (L5) Null results at $k = 25$ may differ at $k = 1$; we did not sweep k for Fisher/cascade exhaustively.

10 Data availability

data/cells/cascade/ (180 npz), data/cells/fisher/ (180 npz), data/results_summaries/mapper_augmentation/ (10-seed output). Master manifest: MANIFEST.sha256.json.

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